SAEON Agile Logbook

Ethan Williams

Field	Record
Student number	221454780
Role	Analyst Researcher and Writer
Project	SAEON Real-Time Rainfall Monitoring System

Week 1 28 July 2026 to 31 July 2026

Tuesday 28 July 2026

Completed since the last stand-up	I reviewed the SAEON monitoring problem and the proposed project scope.
Plan before the next stand-up	I will confirm the component documentation.
Impediments, risks or problems	I need the agreed scope and component choices to remain consistent across the documents.

Wednesday 29 July 2026

Completed since the last stand-up	I researched the selected ESP32, TR-525M and SHT31 components.
Plan before the next stand-up	I will finish the setup and research notes.
Impediments, risks or problems	I need the agreed scope and component choices to remain consistent across the documents.

Thursday 30 July 2026

Completed since the last stand-up	I documented the development setup and evidence requirements.
Plan before the next stand-up	I will review the architecture with the team.
Impediments, risks or problems	I need the agreed scope and component choices to remain consistent across the documents.

Friday 31 July 2026

Completed since the last stand-up	I outlined the sensor-to-cloud architecture and data flow.
Plan before the next stand-up	I will support the firmware and sensor documentation.
Impediments, risks or problems	I need the agreed scope and component choices to remain consistent across the documents.

Week 2 4 August 2026 to 7 August 2026

Tuesday 4 August 2026

Completed since the last stand-up	I worked on the live dashboard simulation and helped plan how the sensor data would be displayed on the dashboard.
Plan before the next stand-up	I will continue testing the available sensors, work on the dashboard mockups, and prepare the demonstration notes.
Impediments, risks or problems	I do not have all the sensors needed for the simulation, which limits the testing I can complete.

Wednesday 5 August 2026

Completed since the last stand-up	I tested a different temperature sensor that also includes humidity and helped with the sensor simulation.
Plan before the next stand-up	I will continue testing the sensors, improve the physical setup using a larger breadboard, and work on ESP32 connectivity.
Impediments, risks or problems	I need a larger breadboard because the current one cannot fit all the components.

Thursday 6 August 2026

Completed since the last stand-up	I tried connecting the HTTP cloud to the ESP32, but the team experienced connection problems. I helped investigate the issue.
Plan before the next stand-up	I will move to Firebase as the cloud server, create a temporary dashboard, and continue working on connecting the ESP32 sensor data to the cloud.
Impediments, risks or problems	I am experiencing problems with the HTTP cloud connection to the ESP32.

Friday 7 August 2026

Completed since the last stand-up	I attended the supervisor meeting and contributed to the discussion about the presentation for the pre-seed fund on 11 August.
Plan before the next stand-up	I will prepare for the upcoming presentation based on the guidance discussed with the supervisor.
Impediments, risks or problems	I have no current impediments.

Week 3 11 August 2026 to 14 August 2026

Planned work area: HiveMQ Cloud Broker Setup
PB03 | US03, US04, US09

Tuesday 11 August 2026

Completed since the last stand-up	I documented the broker setup and connection requirements.

Plan before the next stand-up	I will record the topic permissions.
Impediments, risks or problems	I need the MQTT message format and Firebase records to use consistent identifiers and timestamps.

Wednesday 12 August 2026

Completed since the last stand-up	I recorded the MQTT topic structure and access permissions.
Plan before the next stand-up	I will organise the publish-test evidence.
Impediments, risks or problems	I need the MQTT message format and Firebase records to use consistent identifiers and timestamps.

Thursday 13 August 2026

Completed since the last stand-up	I organised the sample-message delivery records.
Plan before the next stand-up	I will document the storage integration.
Impediments, risks or problems	I need the MQTT message format and Firebase records to use consistent identifiers and timestamps.

Friday 14 August 2026

Completed since the last stand-up	I documented the Firebase storage route and station-linked readings.
Plan before the next stand-up	I will prepare the cellular setup notes.
Impediments, risks or problems	I need the MQTT message format and Firebase records to use consistent identifiers and timestamps.

Week 4 18 August 2026 to 21 August 2026

Tuesday 18 August 2026

Completed since the last stand-up	I drafted the SIM7000G wiring and power notes with the hardware lead.
Plan before the next stand-up	I will confirm the network configuration with the team.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and a stable modem power supply.

Wednesday 19 August 2026

Completed since the last stand-up	I documented the SIM activation, APN and network registration checks.
Plan before the next stand-up	I will organise cellular message examples.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and a stable modem power supply.

Thursday 20 August 2026

Completed since the last stand-up	I organised the available cellular MQTT delivery evidence.
Plan before the next stand-up	I will update the recovery guide.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and a stable modem power supply.

Friday 21 August 2026

Completed since the last stand-up	I updated the cellular troubleshooting and recovery notes.
Plan before the next stand-up	I will review the dashboard requirements.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and a stable modem power supply.

Week 5 25 August 2026 to 28 August 2026

Tuesday 25 August 2026

Completed since the last stand-up	I reviewed the dashboard structure and navigation notes.
Plan before the next stand-up	I will compare live values with sample messages.

Impediments, risks or problems	I need to distinguish live, historical and missing data clearly in the dashboard documentation.

Wednesday 26 August 2026

Completed since the last stand-up	I compared rainfall and temperature displays with the sample messages.
Plan before the next stand-up	I will document the historical filters.
Impediments, risks or problems	I need to distinguish live, historical and missing data clearly in the dashboard documentation.

Thursday 27 August 2026

Completed since the last stand-up	I documented the historical-data filters and displayed units.
Plan before the next stand-up	I will clarify the limits of cloud-command preparation.
Impediments, risks or problems	I need to distinguish live, historical and missing data clearly in the dashboard documentation.

Friday 28 August 2026

Completed since the last stand-up	I recorded the cloud-command preparation as interface work rather than completed remote control.
Plan before the next stand-up	I will prepare the power-system documentation.
Impediments, risks or problems	I need to distinguish live, historical and missing data clearly in the dashboard documentation.

Week 6 1 September 2026 to 4 September 2026

Tuesday 1 September 2026

Completed since the last stand-up	I documented the battery and charge-controller arrangement.
Plan before the next stand-up	I will review the solar setup notes.
Impediments, risks or problems	I need verified power records before describing the system’s operating duration.

Wednesday 2 September 2026

Completed since the last stand-up	I updated the solar integration notes using the team’s setup information.
Plan before the next stand-up	I will organise the alternative-power evidence.

Impediments, risks or problems	I need verified power records before describing the system’s operating duration.

Thursday 3 September 2026

Completed since the last stand-up	I organised the alternative-power test records.
Plan before the next stand-up	I will summarise the power audit.
Impediments, risks or problems	I need verified power records before describing the system’s operating duration.

Friday 4 September 2026

Completed since the last stand-up	I summarised the power-flow audit and its limitations.
Plan before the next stand-up	I will review the station-health documentation.
Impediments, risks or problems	I need verified power records before describing the system’s operating duration.

Week 7 8 September 2026 to 11 September 2026

Tuesday 8 September 2026

Completed since the last stand-up	I reviewed the station-health information and diagnostic labels.
Plan before the next stand-up	I will review the CSV export evidence.
Impediments, risks or problems	I need the dashboard checks and exported values to match the underlying records.

Wednesday 9 September 2026

Completed since the last stand-up	I checked the CSV export notes against the selected monitoring data.
Plan before the next stand-up	I will organise responsive-layout screenshots.
Impediments, risks or problems	I need the dashboard checks and exported values to match the underlying records.

Thursday 10 September 2026

Completed since the last stand-up	I organised phone and desktop interface evidence.
Plan before the next stand-up	I will summarise the pipeline checks.

Impediments, risks or problems	I need the dashboard checks and exported values to match the underlying records.

Friday 11 September 2026

Completed since the last stand-up	I summarised the end-to-end data-pipeline test records.
Plan before the next stand-up	I will document the permanent hardware assembly.
Impediments, risks or problems	I need the dashboard checks and exported values to match the underlying records.

Week 8 15 September 2026 to 18 September 2026

Tuesday 15 September 2026

Completed since the last stand-up	I organised the circuit-transfer diagram and wiring notes.
Plan before the next stand-up	I will update the assembly evidence notes.
Impediments, risks or problems	I need the assembly notes to match the permanent wiring and enclosure.

Wednesday 16 September 2026

Completed since the last stand-up	I updated the assembly documentation using the available soldering records.
Plan before the next stand-up	I will review the enclosure documentation.
Impediments, risks or problems	I need the assembly notes to match the permanent wiring and enclosure.

Thursday 17 September 2026

Completed since the last stand-up	I documented the enclosure preparation and cable- routing arrangement.
Plan before the next stand-up	I will organise the mounting photographs.
Impediments, risks or problems	I need the assembly notes to match the permanent wiring and enclosure.

Friday 18 September 2026

Completed since the last stand-up	I organised the component mounting and sealing photographs.
Plan before the next stand-up	I will prepare the power and calibration audit notes.

Impediments, risks or problems	I need the assembly notes to match the permanent wiring and enclosure.

Week 9 22 September 2026 to 25 September 2026

Tuesday 22 September 2026

Completed since the last stand-up	I summarised the power-drain audit records.
Plan before the next stand-up	I will review solar-charging evidence.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Wednesday 23 September 2026

Completed since the last stand-up	I documented the solar-charging verification results and limitations.
Plan before the next stand-up	I will support deployment documentation.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Thursday 24 September 2026

Completed since the last stand-up	I reviewed the installation preparations while final deployment remained in progress.
Plan before the next stand-up	I will continue recording installation progress.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Friday 25 September 2026

Completed since the last stand-up	I prepared the baseline field-calibration recording sheet; field calibration remained in progress.
Plan before the next stand-up	I will maintain the field-validation evidence templates.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Week 10 29 September 2026 to 2 October 2026

Tuesday 29 September 2026

Completed since the last stand-up	I prepared the evidence checklist for the ongoing 48-hour field stress test.

Plan before the next stand-up	I will update the monitoring log as evidence becomes available.
Impediments, risks or problems	I cannot finalise field-test conclusions while the 48-hour validation remains in progress.

Wednesday 30 September 2026

Completed since the last stand-up	I reviewed the available monitoring notes without treating the field test as complete.
Plan before the next stand-up	I will support the manual comparison records.
Impediments, risks or problems	I cannot finalise field-test conclusions while the 48-hour validation remains in progress.

Thursday 1 October 2026

Completed since the last stand-up	I prepared the manual field-measurement log for rainfall and temperature comparisons.
Plan before the next stand-up	I will continue collecting field evidence with the team.
Impediments, risks or problems	I cannot finalise field-test conclusions while the 48-hour validation remains in progress.

Friday 2 October 2026

Completed since the last stand-up	I drafted the field-audit structure and recorded that final conclusions remained pending.
Plan before the next stand-up	I will keep field conclusions provisional while testing remains in progress.
Impediments, risks or problems	I cannot finalise field-test conclusions while the 48-hour validation remains in progress.

Week 11 6 October 2026 to 9 October 2026

Tuesday 6 October 2026

Completed since the last stand-up	I outlined the presentation around the project problem and prototype.
Plan before the next stand-up	I will update the architecture slides.
Impediments, risks or problems	I need the slides and report to distinguish completed prototype work from open field work.

Wednesday 7 October 2026

Completed since the last stand-up	I updated the architecture diagrams and technical explanations.

Plan before the next stand-up	I will organise dashboard screenshots.
Impediments, risks or problems	I need the slides and report to distinguish completed prototype work from open field work.

Thursday 8 October 2026

Completed since the last stand-up	I organised dashboard screenshots and software slides.
Plan before the next stand-up	I will complete the technical guide.
Impediments, risks or problems	I need the slides and report to distinguish completed prototype work from open field work.

Friday 9 October 2026

Completed since the last stand-up	I consolidated the setup guide, references and technical documentation.
Plan before the next stand-up	I will review the full slide deck.
Impediments, risks or problems	I need the slides and report to distinguish completed prototype work from open field work.

Week 12 13 October 2026 to 16 October 2026

Tuesday 13 October 2026

Completed since the last stand-up	I reviewed the slide deck for clarity and consistency.
Plan before the next stand-up	I will rehearse my assigned section.
Impediments, risks or problems	I need the presentation to describe the demonstrated prototype without claiming final deployment acceptance.

Wednesday 14 October 2026

Completed since the last stand-up	I prepared my speaking notes for the group rehearsal.
Plan before the next stand-up	I will refine the demonstration explanation.
Impediments, risks or problems	I need the presentation to describe the demonstrated prototype without claiming final deployment acceptance.

Thursday 15 October 2026

Completed since the last stand-up	I refined my explanation using the rehearsal structure and review points.
Plan before the next stand-up	I will check the compiled package.
Impediments, risks or problems	I need the presentation to describe the demonstrated prototype without claiming final deployment acceptance.

Friday 16 October 2026

Completed since the last stand-up	I organised the prototype presentation and documentation package.
Plan before the next stand-up	I will review the prototype documentation checklist.
Impediments, risks or problems	I need the presentation to describe the demonstrated prototype without claiming final deployment acceptance.

Week 13 20 October 2026 to 23 October 2026

Tuesday 20 October 2026

Completed since the last stand-up	I checked the setup guide and evidence links for prototype documentation review.
Plan before the next stand-up	I will resolve documentation inconsistencies.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Wednesday 21 October 2026

Completed since the last stand-up	I aligned the documentation with the project’s current scope and task statuses.
Plan before the next stand-up	I will check that field limitations remain clear.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Thursday 22 October 2026

Completed since the last stand-up	I reviewed my presentation notes and the outstanding deployment limitations.
Plan before the next stand-up	I will finalise my prototype handover notes.

Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Friday 23 October 2026

Completed since the last stand-up	I completed my documentation checks and listed the field-deployment actions still in progress.
Plan before the next stand-up	I will update the documentation when deployment and field validation are complete.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.